
Experiment 7: Measurement of Young's Modulus using Strain Gauges

ME-330

Dr. Vibhav Durgesh

vdurgesh@uidaho.edu

Week-08

Contents

1 Learning objective	3
2 Equipment checklist	3
3 Introduction	3
4 Objectives	3
5 Laboratory instruction	4
5.1 Part-1: Build and Test a Half Arm Wheatstone Bridge Circuit	4
5.2 Part-2: Part 2, Preparing for analysis for the Strain and Stress Data.	4
5.3 Part 3, Take Load and Strain Data for the Cantilevered Beam	5
5.4 Part 4, Perform linear regression of stress vs strain to estimate Young's modulus of the beam.	6
5.5 Part 5, Collect uncertainties for each variable used to determine Young's Modulus.	8
5.6 Part 6, Compare uncertainty analysis from two approaches	9
6 Post-Laboratory	10

7	Appendix	11
7.1	Matlab program to acquire data for Strain gauge	11
7.2	Matlab program to plot Stress-strain curve	12

1 Learning objective

The objectives of this experiment are to provide the student with an opportunity to:

- Learn about the strain gauge and Wheatstone bridge
- Learn about the uncertainty propagation and analysis
- Learn about using instrumentation amplifier

2 Equipment checklist

- Cantilevered beam with mounted strain gauges.
- AD622 Instrumentation Amp
- Powered Breadboard & ± 15 Volt source
- Table-top Digital Multi-Meter
- Wire & Resistor Kits
- Screw Driver, Pliers, Wire Stripper
- Precision Matched Resistors
- Calipers
- Measuring Tape
- Weight Set

3 Introduction

Measurement of strain is a fundamental tool used by engineers in both research and design. Strain measurements are often used in systems where force or pressure measurements are required. In some applications the strain (and associated stress) of a part can only be determined by measurement, as calculation is either too complicated or not accurate enough. In this experiment, you will learn to use a metallic strain gauge to measure strain in an end-loaded cantilevered beam. Simultaneously, you will calculate the stress due to bending caused by the end load. You will use these values to create a stress v. strain plot to determine the modulus of elasticity (or Young's modulus) of the beam.

4 Objectives

- Assemble a Wheatstone bridge circuit to measure a signal from two strain gauges affixed to a cantilevered beam.
- Collect enough strain and loading data to perform a linear regression to estimate Young's modulus, E , of the cantilevered beam. Compare the experimental value with published values.

Parts 1-4 first week, Parts 5-6 second week

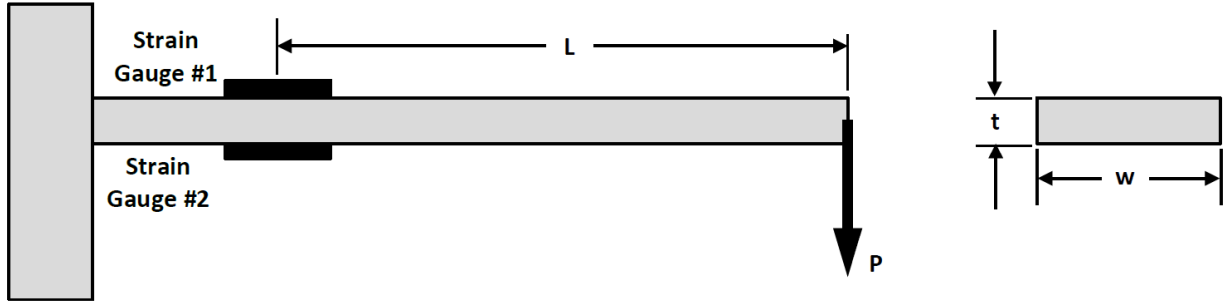


Figure 1: End loaded cantilever beam with two bonded strain gauges.

5 Laboratory instruction

5.1 Part-1: Build and Test a Half Arm Wheatstone Bridge Circuit

- Create a Matlab program to acquire and display the output voltage from the bridge circuit. The DMM will be most useful during circuit debugging.
- On a breadboard, build a half arm Wheatstone bridge circuit as shown in Figure 2. Use the two strain gauges mounted to the cantilevered beam as resistors R#1 and R#2. Use matching precision resistors (120Ω or 350Ω) for R.
- The outputs from the bridge are the inputs to the instrumentation amplifier. Use a 102Ω resistor to set AD622 amplifier gain to $G = 500$. This will amplify the voltage to a range suitable for the table-top multi-meter and acquisition through Matlab. Connect the output of the instrumentation amplifier to the table-top multi-meter and the USB-MyDAQ.
- Make sure to connect the excitation ground to the REF pin on the instrumentation amplifier (2).
- Provide the amplifier and the bridge with a 5V excitation voltage from Ni MyDAQ (2). Use the table-top DMM to measure the amplified output voltage, $V_o \times G$, from the instrumentation amplifier to check that your experimental setup is working properly.

5.2 Part-2: Part 2, Preparing for analysis for the Strain and Stress Data.

- Measure the length between the center of the strain gauge and point of loading, L ,
- Measure the width of the beam, w , and the thickness, t , and
- Estimated all uncertainties all the measured variables.
- Also, determine the local gravity by going to [Local gravity website](#) (use the Latitude value of 46.7324 and elevation of 2579 ft).
- You will be using the general equation $\sigma = My/I$ for stress calculations. The stress equation including all the measured variable is provided in equation 4.

$$\sigma = \frac{My}{I}, \quad (1)$$

$$= \frac{PL(t/2)}{wt^3/12}, \quad (2)$$

$$= \frac{6PL}{wt^3}, \text{ and} \quad (3)$$

$$= \frac{6mgL}{wt^2}. \quad (4)$$

- The measured strain, ϵ , may be found from the half-bridge Wheatstone circuit

$$\frac{\delta V_o}{V_i G} = \frac{GF \epsilon}{2}, \quad (5)$$

$$\epsilon = 2 \frac{\delta V_o}{V_i G} \times \frac{1}{GF} \quad (6)$$

where V_i is excitation voltage, $GF = 2.012$ is the gauge factor, G is the instrumentation amplifier gain, and δV_o is the change in output voltage from loaded to unloaded condition. Use equation (6) for calculating strain values during the experiment script.

Table 1: Table to record your experimental data.

	Mass (kg)	Loaded voltage (v)	Unloaded voltage (v)	$G\delta V_o$	$[G\delta V_o]_{avg}$	ϵ [Use Eq.6]	σ (GPa) [Use Eq.4]
1	0.0						
2	0.4 0.4						
3	0.6 0.6						
4	0.8 0.8						
5	1.0 1.0						
6	1.2 1.2						
7	1.4 1.4						
8	1.6 1.6						
9	1.8 1.8						
10	2.0 2.0						

5.3 Part 3, Take Load and Strain Data for the Cantilevered Beam

- Begin taking load and strain data for the cantilevered beam. First, make a voltage measurement of the beam unloaded. Record voltages acquired from Matlab program.

- Make sure for each measurement you collect 1000 data points at sampling rate of 50 Hz and record the average voltage value in the table provided (Table 1). You have to use the script provided to acquire the data.
- Next, load the beam and record the change in voltage and the amount loaded.
- Lastly, unload the beam and record the voltage again. This will give you two data points for each loading (i.e. the change from unloaded to loaded and the change from loaded to unloaded). Record these values in both your Matlab and your engineering logbooks (for backup!).
- Use the table provided to record your experimental data, with loads in the 1st column (kilograms), unloaded output voltage in the 2nd column, and loaded output voltage in the 3rd column. For each load, create two separate rows in this array. Both rows will have the same load and loaded voltages, but different unloaded voltages (one from before loading, one from after removing the load). During the experiment, repeat these data points in your logbooks to make sure no data is lost! (TAs will be able to explain in detail about the table)
- Repeat the process for at least 9 different loading values. Make sure to spread your measurements across the full range of possible loadings (note that the weights are designed to hang on each other, so heavier loads than 1 kg should be included). The more data you collect, the more accurate your linear regression will be. This will be important to getting accurate results for your report.
- Create a variable “straindata” to store all your strain values and variable “stressdata” to store all stress values.

5.4 Part 4, Perform linear regression of stress vs strain to estimate Young’s modulus of the beam.

- Use the Matlab program provided for curve fitting.
- Calculate stress and strain for all data points, using the polyfit in Matlab to perform a linear regression. The slope of the stress vs. strain plot is Young’s modulus.
- Create a plot of stress (in GPa) vs strain (in micro-strain). Do this after performing the regression, and don’t modify your data.
- To verify your results, check your experimental value for Young’s modulus with published values for steel.

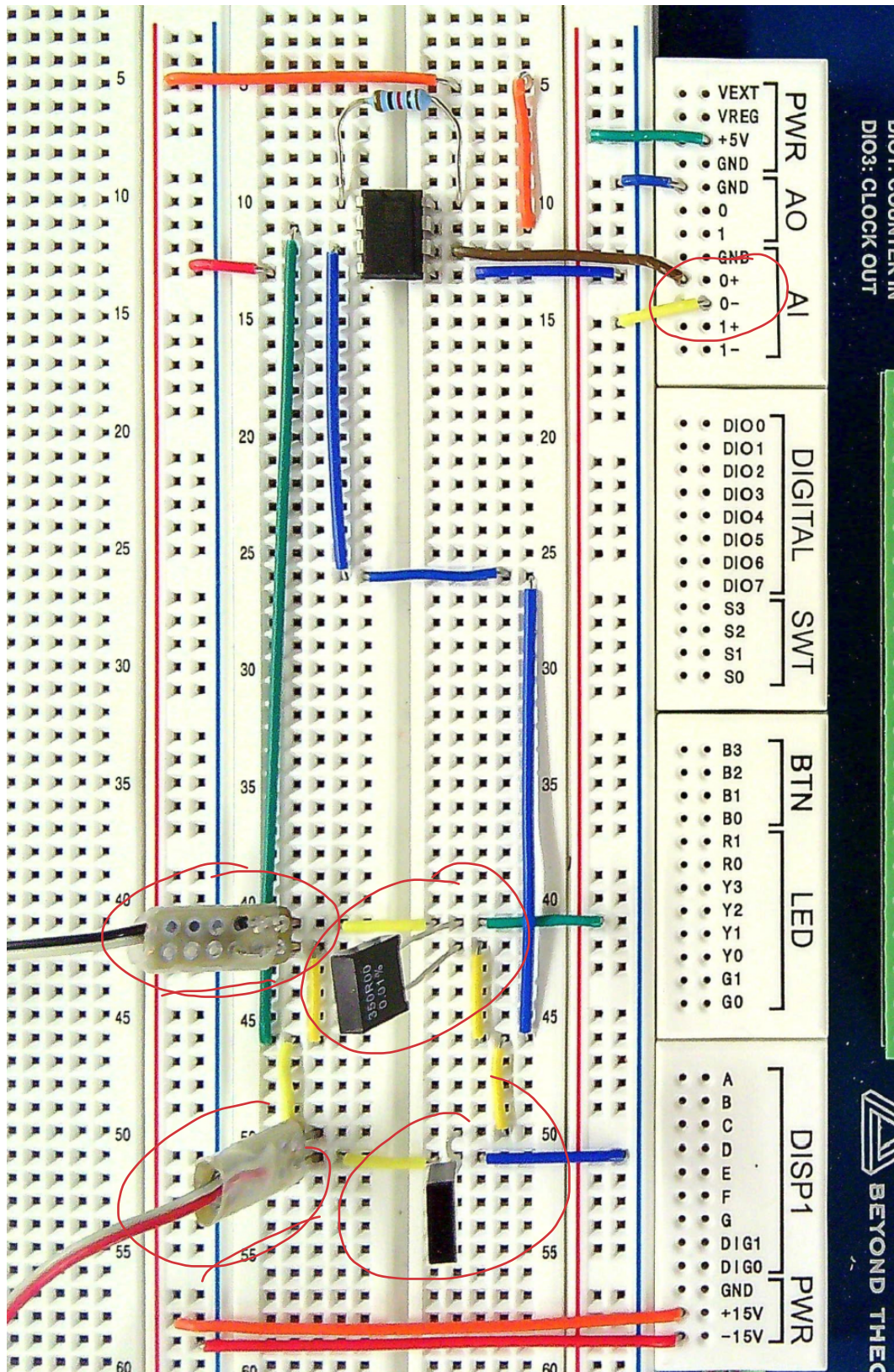


Figure 2: Two strain gauges mounted on a cantilevered beam and a half arm Wheatstone bridge circuit.

Parts 5-6 second week

5.5 Part 5, Collect uncertainties for each variable used to determine Young's Modulus.

- Determine the uncertainty for each variable used to calculate stress and strain in the experiment. Complete the 1st two rows in Table 2. For those measurements performed in lab (e.g. L , w , t), estimate the uncertainty. For the measurement equipment, refer to the manufacturer's datasheets.
- Perform this analysis for three different masses $m = 0.5$ kg, 1.0 kg, and 1.5 kg.
- Use 1% uncertainty in the mass
- Use 0.001% uncertainty in acceleration due to gravity
- Use 1% uncertainty in the GF
- For uncertainty in the op-amp gain (G): use the propagation of error approach with only source of uncertainty is from the resistance used to set the gain on the instrumentation amplifier.
- Uncertainty in the excitation voltage (V_i): resolution of the measurement system used
- Uncertainty in $G\delta V_o$: Uncertainty from resolution of the data acquisition system and first order uncertainty.

Table 2: Table to calculate uncertainty in Young’s modulus using error propagation approach. Fill in the separate tables for m = 0.5, 1.0, and 1.5 kg.

	m (kg)	g (m/s ²)	L(m)	t(m)	w(m)	GF	G	V _i (v)	[δV _o] _{avg} (v)
Variable mean value	0.5								
Vairable uncertainty									
Uncertainty propagated to Young’s modulus (GPa)									
Total uncertainty in Young’s modulus (GPa)									

	m (kg)	g (m/s ²)	L(m)	t(m)	w(m)	GF	G	V _i (v)	[δV _o] _{avg} (v)
Variable mean value	1.0								
Vairable uncertainty									
Uncertainty propagated to Young’s modulus (GPa)									
Total uncertainty in Young’s modulus (GPa)									

	m (kg)	g (m/s ²)	L(m)	t(m)	w(m)	GF	G	V _i (v)	[δV _o] _{avg} (v)
Variable mean value	1.5								
Vairable uncertainty									
Uncertainty propagated to Young’s modulus (GPa)									
Total uncertainty in Young’s modulus (GPa)									

5.6 Part 6, Compare uncertainty analysis from two approaches

- To estimate uncertainty in Young’s Modulus, we will use two approaches
 - The propagation of error approach Use the equation 7.

$$E = \frac{3mgLV_iG(GF)}{wt^2\delta V_o}. \quad (7)$$

- The error from curve fitting approach. The standard error of the slope can be found according to:

$$s_{a1} = s_{yx} \sqrt{\frac{1}{\sum_{i=1}^N (x_i - \bar{x})^2}}, \quad (8)$$

where s_{yx} is the standard error of the the fit.

6 Post-Laboratory

First week: For this experiment, submit the following items on Canvas for your post-laboratory assignment.

- Stress vs. Strain plot with linear regression. Submit a single figure with stress (in MPa) plotted vs strain (in micro- strain). Also include your linear regression line, of which the slope is Young's modulus. Use the text command to place your experimentally determined value (in GPa) for Young's modulus on the plot.
- A straindataVD.m file. Save the files with your initials on the end and submit it on Canvas. Use your initials, not mine!
- Answer all questions in the post-lab assessment on Canvas.

Second week:

- Submit the pdf file with your table for uncertainty propagation.
- Answer all questions in the post-lab assessment on Canvas.

Post-Laboratory Report

This experiment will be the subject of the 2nd of 3 Formal Reports. For this second report, start with the report draft/outline posted on Canvas.

7 Appendix

7.1 Matlab program to acquire data for Strain gauge

```
1 % This is a genric program to acquire the data from NI- 6001
2 %% Sample data acquisition program for calibration purpose it logs the data in a
3 % file for later use
4 % Date: July 19th
5 % Dr. Vibhav Durgesh
6 % Rev 0.0
7 % This is basic program to acquire data from a data acquisition device.
8 % User has to provide appropriate information see - begining of the code
9 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
10 % THIS CODE OVERWRITE THE DATA. PLEASE MOVE THE FILES OR REMANE PRIOR TO
11 % RERUNNING THE CODE
12 % MAKE SURE TO SELECT CORRECT CHANNEL FOR CALIBRATION
13 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
14
15 clear all
16 close all
17 clc
18
19 %% ----- START USER REQUIRED INFORMATION ----- %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
20 Fs = 50; %Sampling rate Data per sec
21 T = 10; % Time to acquire data
22 filename = '..\Data\SG_Data_1500gms.dat'; % To store strain gauge data file
23 fid1 = fopen(filename,'w');
24 %% -----END USER REQUIRED INFORMATION ----- %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
25 d = daq.getDevices;
26 s = daq.createSession('ni');
27 addAnalogInputChannel(s,'MyDaq1', [0], 'Voltage');
28 s.Rate = Fs; % Sampling rate modify as required
29 s.DurationInSeconds = T; % Sampling time modify as required
30 lh1 = addlistener(s,'DataAvailable', @(src, event)plotAndLogData(src, event));
31 lh2 = addlistener(s,'DataAvailable', @(src, event)logData(src, event, fid1));
32 errorListener = addlistener(s, 'ErrorOccurred', @(src,event) disp(getReport(event.Error)));
33 drawnow
34 s.IsContinuous = true;
35 startBackground(s);
36 pause(T)
37 delete(s)
38 delete(lh1)
39 delete(lh2)
40 fclose(fid1)
41 data = readmatrix(filename);
42 avgVolt = mean(data(:,2));
43 gcf;
44 title(['Average voltage: ' sprintf('%3.4f',avgVolt) ' (v)'], 'fontname','times', 'fontsize'
45 ,14)
46
47 function plotAndLogData(src,event)
48 time = event.TimeStamps;
49 voltage = event.Data; %Potentiometer data in voltage
50 figure(1)
51 plot(time, voltage, 'k.-');hold on
52 xlabel('time (s)')
53 ylabel('Voltage (v)')
54 end
55
56 function logData(src, evt, fid)
57 % Add the time stamp and the data values to data. To write data sequentially,
58 % transpose the matrix.
59 data = [evt.TimeStamps evt.Data] ;
60 fprintf(fid, '%f \t %f \n', data');
```

7.2 Matlab program to plot Stress-strain curve

```

1 % This is a program to generate strain-stress plot
2 % Date: July 19th
3 % Dr. Vibhav Durgesh
4 % Rev 0.0
5 % This is basic program to demonstrating use of polyfit command and
6 % calculating norm and standard error of fit.
7 % User has to provide appropriate information see - beginning of the code
8 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
9
10 clear all
11 close all
12 clc
13 %%% USER HAVE TO MANUALLY ADD THIS INFORMATION %%%%%%%%%
14 %%%
15
16 mass = [0 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0]; %Change to appropriate weight
17 Gdvo_avg = [0 0.1447 0.2174 0.2924 0.3628 0.4342 0.5081 0.5809 0.6521
18 00.7281]; % Average voltage output
19 L = 36/100; % length of beam using a tape measure
20 w = 1.499*(2.54/100); %width of the beam using a caliper
21 t = 0.2515*(2.54/100); %thickness of the beam using a micrometer
22 g = 9.8051; % gravitational constant (m/s)
23 GF = 2.012; % Gauge Factor
24 G = 497.7538855; % gain using Rg = 101.66 ohm
25 Vi = 10; %Excitation Voltage to the wheatstone bridge circuit
26
27 m = mass; % mass (kg)
28
29 %% Calculating Stress
30 stressdata = ((6*m*g*L)/(w*t^2))/(10^9); %Stress(MPa)
31
32 %% Calculating Strain
33 straindata = (2*Gdvo_avg)/(Vi*G*GF); %strain (unitless)
34
35 t_nuP = 2.110; %Value read from t-student's table (19 data points)
36 plotTitle = 'First Name Last Name's plot';
37 %%% END OF USER INFORMATION %%%%%%%%%
38 [p,s] = polyfit(straindata,stressdata,1); % Curve fitting data with 1st order fit
39 xfit = straindata;% Generating x-data for curve fitting
40 yfit = polyval(p,xfit);
41 nu = s.df; %Getting degree of freedom for the curve fit
42 norm = s.normr; % Getting the norm of the curve fitting
43 syx = norm/sqrt(nu);
44
45 % Generating figure with specific size
46 figure(1)
47 set(gcf,'unit','inches','position',[0.50 0.50 6.50 3.50],...
48 'defaultaxesfontsize',10,'defaultaxesfontname','times');
49 % Plotting data
50 plot(xfit,yfit,'b-','linewidth',2); hold on
51 plot(straindata,stressdata,'ro','markersize',9,'markerfacecolor','r')
52 xlabel('\epsilon (m/m)')
53 ylabel('\sigma (GPa)')
54 grid on
55 % Now adding 95% confidence level in the plot
56 y_cl_low = yfit - t_nuP*syx; % Using regression analysis
57 y_cl_up = yfit + t_nuP*syx;
58 plot(xfit,y_cl_low,'--','color',[0.2 0.2 0.2],'HandleVisibility','off');
59 plot(xfit,y_cl_up,'--','color',[0.2 0.2 0.2]);
60
61 legend('Curve fit','Expt. data','95% CL range','location','Northwest')
62 title(plotTitle)
63 text(1e-4,0.01,sprintf('%s = %3.4f%s+%3.4f','\sigma',p(1),'\epsilon',p(2)),'Fontname','times')
64 text(1e-4,0.005,sprintf('Norm = %3.4f',s.normr),'Fontname','times')

```

```

65 text(1e-4,0,sprintf('s_{yx} = %3.8f%s',syx,'GPa'),'Fontname','times')
66 % % Saving the files in png and pdf format
67 figName = ['..\Figures\FirstnameLastname'];
68 set(gcf,'PaperPositionMode','auto')
69 print(figName,'-dpng','-r600')
70 set(gcf,'PaperUnits','inches','Units','inches');
71 figpos = get(gcf,'Position');
72 set(gcf,'PaperSize',figpos(3:4),'Units','inches');
73 print(figName,'-dpdf','-r600')
74 %

```